

Danish Vasdev

Third Year Undergraduate
Major: Electrical Engineering

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ACADEMIC QUALIFICATIONS

Year	Educational Qualification	Institute	Performance (CPI / %)
2021 - 2025	BTech.	Indian Institute of Technology Kanpur	8.42/10.00
2021	Class XII (CBSE)	Army Public School, Pune	97.4%
2019	Class X (CBSE)	Army Public School, Pune	94.6%

SCHOLASTIC ACHIEVEMENTS

- Received **Academic Excellence award** for exceptional academic performance in the year 2021-22 at IIT Kanpur
- Achieved All India Rank **942** in JEE Advanced 2021, All India Rank **3595** in JEE Main 2021 among 1.5 million+ appearing students
- Received certificate of merit for being **amongst top 1%** in Indian Olympiad Qualifier in Chemistry, and Physics 2020-21
- Recipient of the prestigious **KVPY Fellowship Award** by IISc Bangalore, achieving an **All India Rank of 131** in the year 2020
- Qualified the **National Standard Examination in Astronomy 2019**, conducted by Homi Bhabha Centre for Science Education
- Stood **amongst the top 40 out of 600+ participants** in The Indian National Astronomy Olympiad (INAO) Jr Pool – 2019
- Attended the **International Astronomy Olympiad (IraO)** Orientation cum Selection Camp at Regional Science Centre, Guwahati

PROFESSIONAL WORK EXPERIENCE

Intern – Saras M&E LLC, Dubai

Objective	● Identify and suggest a solution for smoothening business operations of the transformer maintenance service
Approach	● Analysed traditional method of preventive maintenance using tests such as SFRA test, time delta test etc ● Researched the implementation of predictive maintenance using Dissolved Gas Analysis in oil transformers ● Developed a LSTM model to predict transformer health using historical DGA data of transformers at various sites ● Performed cost analysis for predictive vs preventive maintenance, cutting costs by minimizing maintenance
Result	● Shift from preventive to predictive maintenance showed productivity increase by predicting transformer health

PROJECT WORK

Azure Cognitive Services for Accountants- Dr Arvind Sathi, Director Data Scientist, KPMG LLP

Objective	● To design a course on use cases of Computer vision and NLP for accounting professionals using Azure and GPT
Approach	● Used Azure form recognizer API to design various document information extraction models for tabular data ● Interfaced the application with GPT-3.5 API to provide a user-friendly natural language based query system ● Created information extraction model on Nutrition labels which can be queried directly by chatting with the app
Result	● Developed real world based working examples for the course using Azure form recognizer and GPT-3.5 API

3D Visualization – Prof S Dutta (CSE dept, IIT Kanpur)

Objective	● Enhance explainability of latent features , model confidence of ML model using graphical visualizations
Approach	● Reviewed current state of the art papers on visual analytics of ML models for representing model parameters ● Built interactive web interface for exploring model training process for MLP, CNN models using d3.js framework ● Developed visualization for GAN model confidence using Monte Carlo sampling on 3d reconstruction data
Result	● Built an interactive web platform to analyse various model parameters like accuracy, model bias, confidence

RL based Racing – Deep Racer Competition –International event hosted by AWS

Objective	● Harness reinforcement learning techniques to optimize reward function of model for racing a virtual car
Approach	● Reviewed winning strategies , papers on Deep racer used by research teams, previous international winners ● Designed 20+ different reward functions emphasizing on various strategies , improving the result on each step ● Trained the model on a reward function with enhanced features such as corner cutting, path optimization
Result	● Stood 29th amongst 1200 participants internationally in the AWS deep racer student league competition

POSITIONS OF RESPONSIBILITY

Team Head, VISION IITK

Leadership	● Led a team of 15 students in training and participation for various competitions in autonomous ground navigation
Initiatives	● Organized and prepared the team for international competitions such as the ICRA BARN challenge, ICRA F1/10 ● Revamped the team's marketing strategy by introducing a dedicated marketing team for approaching sponsors ● Reintroduced hardware-based competitions within the team, aligning with the team's pre-COVID era initiatives
Impact	● Revitalised the team by establishing stronger roots and set the momentum for future growth and exploration

Science and Technology Secretary - Hall Executive Committee (Hall 5), IIT, Kanpur

Leadership	● Directed the efforts of the Nawabs pool team in Takneek 22, the inter hall science competition, across 10+ problem statements from all domains of the Science and technology council in a competition spanning more than 20 days
Initiatives	● Created and coordinated a team of more than 200 students across batches, working together in various problems ● Oversaw the training of over 100 freshers, creating a medium for effective knowledge transfer from senior batches ● Managed and allocated a budget of over 40,000 to acquire equipment for participation in various competitions
Impact	● Secured the trophy of Takneek 22, amongst 5 pools, consisting of a total of 1000+ students across all batches

Company Coordinator - Students Placement Office

Initiatives	● Facilitated the placement process by liaising and facilitating company visits, managing test and interview logistics ● Handled senior recruits of leading firms and served as POC for companies including Barclays, Intel, Devrev.ai .
Impact	● Assisted in securing internship of 450+ third year students as well as placement of 1200+ final year students

SOCIAL AND EXTRACURRICULAR WORK

- Student Guide** - Counselling Service, IIT Kanpur - Mentored 5 first-year students, guided them emotionally and academically